

PROFORMA — Standard Web Application Infrastructure Billing

LIONGATE SARL — Infrastructure Cost Estimation

Document: PROFORMA-02

Context: Standard Web Application (Backend API + Frontend + Database)

Hosting Provider: Hetzner Cloud

Currency: EUR (€)

Version: 1.0

How to read this document:

This proforma covers all infrastructure costs to deploy and operate a standard web application (e.g., NestJS API + Next.js frontend + PostgreSQL) across Development, Staging, and Production environments. Four tiers are presented from minimum viable to high-availability.

Reference Stack

Layer	Technology	Notes
Backend	NestJS (TypeScript) or FastAPI (Python)	Single API service
Frontend	Next.js (React)	SSR or static export
Database	PostgreSQL 16	Primary data store
Cache	Redis (optional)	Sessions, queues
Reverse Proxy	Nginx	SSL termination
Container	Docker + Docker Compose	All services
CI/CD	GitHub Actions	Build + deploy
SSL	Let's Encrypt	Free, auto-renew

Tier 1 — Minimum Viable Setup

App + DB on a single server. For low-traffic apps, MVPs, or early-stage client projects.

Traffic assumption: < 1,000 requests/day, < 100 concurrent users.

Environment: Production (Minimum)

#	Resource	Spec	Unit Price	Qty	Monthly (€)	Annual (€)
1	Cloud Server (App + DB)	CX32 — 4 vCPU, 8 GB RAM, 80 GB NVMe	€8.21	1	8.21	98.52
2	Hetzner Volume (DB + uploads)	50 GB block storage	€2.60	1	2.60	31.20
3	Storage Box (Backups)	BX11 — 100 GB	€3.43	1	3.43	41.16
4	Floating IP	Static IPv4	€2.00	1	2.00	24.00
5	SSL Certificate	Let's Encrypt	Free	1	0.00	0.00
6	Domain (amortised)	.com annual registration	€15.00/yr	1	1.25	15.00
7	Outbound email	Brevo Free — 300/day	Free	1	0.00	0.00
8	External uptime monitoring	UptimeRobot Free	Free	1	0.00	0.00
—	SUBTOTAL				€17.49	€209.88

Environment: Staging

#	Resource	Spec	Monthly (€)	Annual (€)
1	Cloud Server (App + DB)	CX22 — 2 vCPU, 4 GB RAM	4.35	52.20
2	Floating IP	Static IPv4	2.00	24.00

#	Resource	Spec	Monthly (€)	Annual (€)
3	SSL	Let's Encrypt	0.00	0.00
—	SUBTOTAL		€6.35	€76.20

💡 Stop staging server off-hours → effective cost ≈ **€2.54/mo**

Environment: Development

#	Resource	Spec	Monthly (€)	Annual (€)
1	Local Docker Compose	Developer workstation	0.00	0.00
2	Shared dev VPS (optional)	CX22 — team-shared	4.35	52.20
—	SUBTOTAL		€0.00 – €4.35	€0.00 – €52.20

Tier 1 — Total Summary

Environment	Monthly MIN (€)	Monthly MAX (€)	Annual MIN (€)	Annual MAX (€)
Production	17.49	20.11	209.88	241.32
Staging	2.54 <i>(paused)</i>	6.35	30.48	76.20
Development	0.00	4.35	0.00	52.20
TOTAL TIER 1	€20.03	€30.81	€240.36	€369.72

Tier 2 — Recommended Setup

App and database on separate servers. Standard production-grade for most client projects.
Traffic assumption: 1,000–10,000 requests/day, up to 300 concurrent users.

Environment: Production (Recommended)

#	Resource	Spec	Unit Price	Qty	Monthly (€)	Annual (€)
1	App Server (API + Frontend)	CX42 — 8 vCPU, 16 GB RAM	€17.77	1	17.77	213.24
2	DB Server	CX32 — 4 vCPU, 8 GB RAM	€8.21	1	8.21	98.52
3	Volume — DB data	100 GB — attached to DB server	€5.20	1	5.20	62.40
4	Volume — Media / Uploads	50 GB — attached to App server	€2.60	1	2.60	31.20
5	Private Network	App ↔ DB VLAN	Free	1	0.00	0.00
6	Storage Box (Backups)	BX31 — 1 TB	€8.69	1	8.69	104.28
7	Floating IP	Static IPv4	€2.00	1	2.00	24.00
8	SSL Certificate	Let's Encrypt	Free	1	0.00	0.00
9	Domain (amortised)	Annual registration	€15.00/yr	1	1.25	15.00
10	Outbound email	Brevo Starter — 20k/mo	€19.00	1	19.00	228.00
11	External uptime monitoring	UptimeRobot Free	Free	1	0.00	0.00
12	Object Storage (media CDN fallback)	S3 — 20 GB avg	€0.24	1	0.24	2.88
—	SUBTOTAL				€64.96	€779.52

Environment: Staging

#	Resource	Spec	Monthly (€)	Annual (€)
1	Cloud Server (App + DB)	CX32	8.21	98.52
2	Volume	50 GB	2.60	31.20
3	Floating IP	Static IPv4	2.00	24.00
4	SSL	Let's Encrypt	0.00	0.00
—	SUBTOTAL		€12.81	€153.72

Environment: Development

#	Resource	Spec	Monthly (€)	Annual (€)
1	Local Docker Compose	Developer workstation	0.00	0.00
2	Shared dev VPS (optional)	CX22	4.35	52.20
—	SUBTOTAL		€0.00 – €4.35	€0.00 – €52.20

Tier 2 — Total Summary

Environment	Monthly MIN (€)	Monthly MAX (€)	Annual MIN (€)	Annual MAX (€)
Production	64.96	74.70	779.52	896.40
Staging	5.12 <i>(paused)</i>	12.81	61.44	153.72
Development	0.00	4.35	0.00	52.20
TOTAL TIER 2	€70.08	€91.86	€840.96	€1,102.32

Tier 3 — Full Production Setup

Full observability, Redis cache, dedicated monitoring, production hardening.

Traffic assumption: 10,000–50,000 requests/day, up to 1,000 concurrent users.

Environment: Production (Full)

#	Resource	Spec	Unit Price	Qty	Monthly (€)	Annual (€)
1	App Server	CX42 — 8 vCPU, 16 GB RAM	€17.77	1	17.77	213.24
2	DB Server	CX42 — 8 vCPU, 16 GB RAM	€17.77	1	17.77	213.24
3	Redis Server	CX22 — 2 vCPU, 4 GB RAM	€4.35	1	4.35	52.20
4	Volume — DB data	200 GB	€10.40	1	10.40	124.80
5	Volume — Media / Uploads	100 GB	€5.20	1	5.20	62.40
6	Private Network	VLAN all servers	Free	1	0.00	0.00
7	Storage Box (Backups)	BX31 — 1 TB	€8.69	1	8.69	104.28
8	Object Storage (media)	S3 — 50 GB avg	€0.60	1	0.60	7.20
9	Floating IP	Static IPv4	€2.00	1	2.00	24.00
10	Monitoring Server (shared)	CX22 — Grafana + Prometheus + Loki	€4.35	1	4.35	52.20
11	SSL Certificate	Let's Encrypt	Free	1	0.00	0.00
12	Domain (amortised)	Annual	€15.00/yr	1	1.25	15.00
13	Outbound email	Brevo Starter — 20k/mo	€19.00	1	19.00	228.00
14	External uptime monitoring	UptimeRobot Pro	€7.00	1	7.00	84.00

#	Resource	Spec	Unit Price	Qty	Monthly (€)	Annual (€)
15	VM Snapshot (weekly, 2 kept)	~40 GB avg	€0.48	1	0.48	5.76
—	SUBTOTAL				€98.86	€1,186.32

Environment: Staging

#	Resource	Spec	Monthly (€)	Annual (€)
1	Cloud Server	CX32 — app + DB	8.21	98.52
2	Volume	50 GB	2.60	31.20
3	Floating IP	Static IPv4	2.00	24.00
—	SUBTOTAL		€12.81	€153.72

Tier 3 — Total Summary

Environment	Monthly MIN (€)	Monthly MAX (€)	Annual MIN (€)	Annual MAX (€)
Production	98.86	113.69	1,186.32	1,364.28
Staging	5.12 <i>(paused)</i>	12.81	61.44	153.72
Development	0.00	4.35	0.00	52.20
TOTAL TIER 3	€103.98	€130.85	€1,247.76	€1,570.20

Tier 4 — High-Availability Setup

Load-balanced multi-instance app, dedicated CPU DB, CDN, full HA configuration.

Traffic assumption: 50,000+ requests/day, 2,000+ concurrent users.

Environment: Production (HA)

#	Resource	Spec	Unit Price	Qty	Monthly (€)	Annual (€)
1	App Servers (×2, behind LB)	CX42 — 8 vCPU, 16 GB RAM	€17.77	2	35.54	426.48
2	DB Server (dedicated CPU)	CCX23 — 4 vCPU dedicated, 16 GB RAM	€22.75	1	22.75	273.00
3	Redis Server	CX32 — 4 vCPU, 8 GB RAM	€8.21	1	8.21	98.52
4	Load Balancer	LB21 — up to 25 targets	€10.79	1	10.79	129.48
5	Volume — DB data	300 GB	€15.60	1	15.60	187.20
6	Volume — Shared uploads (×2 apps)	200 GB each	€10.40	2	20.80	249.60
7	Private Network	VLAN	Free	1	0.00	0.00
8	Storage Box (Backups)	BX31 — 1 TB	€8.69	1	8.69	104.28
9	Object Storage (media + CDN origin)	S3 — 100 GB avg	€1.20	1	1.20	14.40
10	Floating IP × 2	Static IPv4 (app + LB)	€2.00	2	4.00	48.00
11	Monitoring Server (dedicated)	CX32 — Grafana full stack	€8.21	1	8.21	98.52
12	SSL Certificate	Let's Encrypt wildcard	Free	1	0.00	0.00
13	Cloudflare Pro (CDN + WAF)	DDoS + proxy	€20.00	1	20.00	240.00

#	Resource	Spec	Unit Price	Qty	Monthly (€)	Annual (€)
14	Domain (amortised)	Annual	€15.00/yr	1	1.25	15.00
15	Outbound email	Brevo Business — 100k/mo	€49.00	1	49.00	588.00
16	External uptime monitoring	UptimeRobot Pro	€7.00	1	7.00	84.00
17	VM Snapshots (rotation ×4)	~80 GB avg	€0.95	1	0.95	11.40
—	SUBTOTAL				€213.99	€2,567.88

Tier 4 — Total Summary

Environment	Monthly MIN (€)	Monthly MAX (€)	Annual MIN (€)	Annual MAX (€)
Production	213.99	246.09	2,567.88	2,953.08
Staging	5.12 <i>(paused)</i>	12.81	61.44	153.72
Development	0.00	4.35	0.00	52.20
TOTAL TIER 4	€219.11	€263.25	€2,629.32	€3,159.00

Standard Web App — Tier Comparison Dashboard

Tier	Setup Type	Monthly MIN	Monthly MAX	Annual MIN	Annual MAX	Best for
T1	MVP — all-in-one CX32	€20.03	€30.81	€240	€370	Prototype, MVP
T2	Recommended — split	€70.08	€91.86	€841	€1,102	Standard client project

Tier	Setup Type	Monthly MIN	Monthly MAX	Annual MIN	Annual MAX	Best for
T3	Full production + Redis	€103.98	€130.85	€1,248	€1,570	Active SaaS product
T4	High-availability, LB	€219.11	€263.25	€2,629	€3,159	High-traffic platform

Hidden & Variable Costs

Cost Item	Frequency	Estimated Amount	Notes
Domain renewal	Annual	€10–€30	Per domain
Snapshot storage growth	Monthly	+€0.50–€3.00	Grows with server size
Additional subdomains	Annual	Included in domain	Same registrar zone
DB read replica (future)	Monthly	+€8.21–€22.75	For read-heavy scaling
Redis cluster (future)	Monthly	+€8.21–€17.77	For high-availability cache
Commercial SSL (if required)	Annual	€50–€300	Only if client mandates
Cloudflare Workers / Pages	Monthly	€0–€20	For edge compute or static hosting
GitHub Actions CI minutes	Monthly	€0 (free tier) or €0.008/min	2,000 free min/mo
Container registry storage	Monthly	€0–€5	GHCR included with GitHub

Budget Planning Summary (Standard Web App)

LIONGATE SARL – Standard Web App Budget Reference

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MINIMUM MONTHLY BUDGET:	€20.03	(Tier 1 – MVP / all-in-one)
RECOMMENDED MONTHLY BUDGET:	€70.08	(Tier 2 – split app+db)
FULL PRODUCTION BUDGET:	€103.98	(Tier 3 – Redis + monitoring)
HIGH-AVAILABILITY BUDGET:	€219.11	(Tier 4 – LB + multi-app)

MINIMUM ANNUAL BUDGET:	€240.36
RECOMMENDED ANNUAL BUDGET:	€840.96
FULL PRODUCTION ANNUAL:	€1,247.76
HIGH-AVAILABILITY ANNUAL:	€2,629.32

CONTINGENCY BUFFER (+15%):

Tier 2 recommended: $€840.96 \times 1.15 = €967.10/\text{yr}$

Tier 3 full: $€1,247.76 \times 1.15 = €1,434.92/\text{yr}$

Document: PROFORMA-02 | Context: Standard Web Application

Prices: Hetzner Cloud published rates + market SMTP/monitoring rates

Review: Quarterly or on Hetzner pricing change

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